

When is $\binom{n}{2}$ a product of consecutive primes?

A guided introduction to Erdős Problem #386 (case $k = 2$)

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Abstract

This is an expository companion to the research note *Binomial coefficients $\binom{n}{2}$ that are products of consecutive primes*. It explains, from first principles, every idea used there. The ideas are: why $\binom{n}{2}$ splits into two nearly balanced pieces (an observation due to StijnC on the Erdős Problems forum); an explicit “balance inequality”; how an explicit prime-gap theorem of Dusart turns that inequality into concrete bounds; how the computer search is organized and checked; what a Lean formalization certifies; and why a full solution appears out of reach. Worked examples and exercises (with solutions) are included. Nothing here solves the problem.

Contents

1	The problem	1
1.1	A test for “is this a binomial coefficient?”	2
2	The split into two teams	2
3	The balance inequality	2
4	From intuition to proof: Dusart’s prime-gap theorem	3
5	The computer search	3
6	The results	4
7	What Lean certifies	4
8	Why the full problem looks hard	5
9	Exercises	5

1 The problem

The binomial coefficient $\binom{n}{2} = n(n-1)/2$ counts the ways to choose 2 objects from n . A *block* is a product of consecutive primes with nothing skipped. For example, $3 \cdot 5 \cdot 7$ is a block, but $7 \cdot 13$ is not, because it skips 11.

n	$\binom{n}{2}$	block
4	6	$2 \cdot 3$
6	15	$3 \cdot 5$
15	105	$3 \cdot 5 \cdot 7$
21	210	$2 \cdot 3 \cdot 5 \cdot 7$
715	255255	$3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$

Question (Erdős–Graham). Are there infinitely many n for which $\binom{n}{2}$ is a block?

The answer is widely believed to be *no*, but no proof is known. This document explains what *can* be proved, and how far a careful computation reaches.

1.1 A test for “is this a binomial coefficient?”

Given a number P , is there an n with $\binom{n}{2} = P$? Multiply by 8 and add 1:

$$\frac{n(n-1)}{2} = P \iff 4n^2 - 4n + 1 = 8P + 1 \iff (2n-1)^2 = 8P + 1.$$

So P is of the form $\binom{n}{2}$ exactly when $8P + 1$ is a perfect square s^2 , and then $n = (1 + s)/2$.

Example 1. $P = 105$: $8 \cdot 105 + 1 = 841 = 29^2$, so $n = (1 + 29)/2 = 15$. $P = 3 \cdot 5 \cdot 7 \cdot 11 = 1155$: $8 \cdot 1155 + 1 = 9241$, and $96^2 = 9216 < 9241 < 9409 = 97^2$, so 1155 is not of the form $\binom{n}{2}$.

2 The split into two teams

Two facts about consecutive integers do all the work.

- (i) n and $n - 1$ share no prime factor: a common factor would divide their difference, 1.
- (ii) Exactly one of $n, n - 1$ is even.

Call the even one E and the odd one O , and put $A = E/2$. Then

$$\binom{n}{2} = A \cdot O, \quad O = 2A + 1 \text{ or } O = 2A - 1.$$

If $\binom{n}{2}$ is a block, each prime of the block divides exactly one of A and O . So the block’s primes split into two *teams*: S (the primes of A) and T (the primes of O), with

$$\left| 2 \prod S - \prod T \right| = 1.$$

In words: **one team’s product is almost exactly twice the other’s**. This observation is due to StijnC on the Erdős Problems forum [2].

Example 2 ($n = 715$). $O = 715 = 5 \cdot 11 \cdot 13$ and $A = 714/2 = 357 = 3 \cdot 7 \cdot 17$. So $T = \{5, 11, 13\}$, $S = \{3, 7, 17\}$, and $2 \cdot 357 = 714 = 715 - 1$.

3 The balance inequality

Lemma 3 (Balance inequality). *Let $n \geq 4$, and suppose every prime factor of $\binom{n}{2}$ lies in $[p, r]$ with $p \geq 5$. Let k be the number of prime factors (with multiplicity) and $h = \lfloor k/2 \rfloor$. Then*

$$\left(\frac{r}{p}\right)^h \geq 2 - \frac{1}{p}.$$

Intuition. All team members lie between p and r . If the primes are packed very closely (r/p near 1), then any two teams have products that are either about equal or differ by a huge factor. They can never differ by a factor of almost exactly 2. The inequality measures how much “spread” is needed.

Proof. Let $s = |S|$ and $t = |T|$ (counted with multiplicity), so $s + t = k$, with $p^s \leq A \leq r^s$ and $p^t \leq O \leq r^t$. Since $n \geq 4$, $A \geq 2$ and $O \geq 3$, so $s, t \geq 1$.

Case $s = t$. $O/A = 2 \pm 1/A \geq 2 - 1/p$, while $O/A \leq r^s/p^s$.

Case $s < t$. $p^{s+1} \leq p^t \leq O \leq 2A + 1 \leq 2r^s + 1$, so $(r/p)^s \geq p/2 - 1/(2p) \geq 2 - 1/p$ (because $p^2 - 4p + 1 \geq 0$).

Case $s > t$. $2p^{t+1} \leq 2A \leq O + 1 \leq r^t + 1$, so $(r/p)^t \geq 2p - p^{-t} \geq 2 - 1/p$.

In every case $(r/p)^{\min(s,t)} \geq 2 - 1/p$. Since $\min(s,t) \leq h$ and $r/p \geq 1$, the lemma follows. $\square$

Remark 4 (Sharpness). For $n = 14$: $\binom{14}{2} = 91 = 7 \cdot 13$, $p = 7$, $r = 13$, $k = 2$, and $13/7 = 2 - 1/7$ exactly. The inequality therefore cannot be improved. But $7 \cdot 13$ skips 11, so this is not a solution to the problem.

Why large starting primes force long blocks. Near 10^{10} , consecutive primes are on average about $\ln(10^{10}) \approx 23$ apart. Even 1000 consecutive primes there span a ratio $r/p \approx 1 + 23000/10^{10} \approx 1.0000023$. Raising that to the power 500 gives about 1.001, nowhere near 2. To reach 2 one needs tens of thousands of primes, and multiplying that many primes near 10^{10} gives an astronomically large number.

4 From intuition to proof: Dusart’s prime-gap theorem

“Primes near x are closely packed” needs a theorem with explicit constants. We use:

Theorem 5 (Dusart [1, Cor. 5.5]). *For all $x \geq 468\,991\,632$ there is a prime q with $x < q \leq x \left(1 + \frac{1}{5000 \ln^2 x}\right)$.*

So when all primes of a block are at least $X_0 = 10\,726\,905\,041$, each step to the next prime multiplies by at most $1 + \varepsilon$, where $\varepsilon = 1/(5000 \ln^2 X_0) \approx 3.75 \times 10^{-7}$. Over k primes,

$$\frac{r}{p} \leq (1 + \varepsilon)^{k-1}, \quad \text{so} \quad \left(\frac{r}{p}\right)^{\lfloor k/2 \rfloor} \leq e^{\varepsilon(k-1)\lfloor k/2 \rfloor}.$$

Lemma 3 needs this to be at least $2 - 1/p \approx 2$, so $\varepsilon(k-1)\lfloor k/2 \rfloor \geq \ln 2$, roughly $k^2 \geq 1.386/\varepsilon \approx 3.7 \times 10^6$. The exact computation gives $k \geq 1924$.

5 The computer search

Starting primes below X_0 are handled by computer. There are exactly $\pi(X_0) = 486\,570\,088$ of them.

Step 1: cheap exclusion. For most starting primes p , a simple integer inequality ($100h(r-p) < 68p$, which implies $(r/p)^h < 2 - 1/p$ because $(1+x)^h \leq e^{hx}$ and $e^{0.68} < 1.975$) shows that Lemma 3 fails for *every* block length that matters. The start is then discarded in constant time.

Step 2: full check. For the remaining few million starts, every block length is tested with the square test of §1.1.

The filter trick. The block products are far too large to write down, since they have thousands of digits. But if $8P+1$ is a square, it is also a square modulo any prime ℓ . About half of all residues mod ℓ are non-squares, so each “filter prime” ℓ rejects about half of the impostors. With 48 filters, an impostor survives with probability about 2^{-48} . Any survivor is re-checked exactly with integer square roots. In every run, the only survivors were the five known solutions.

Checks on the checker.

- Three independent programs agree for $n \leq 10^9$.
- The number of starting primes processed equals $\pi(X_0)$, computed by an independent program. That program was itself validated against the known values $\pi(10^6)$, $\pi(10^8)$ and $\pi(10^{10})$.
- The filter code was tested against Euler’s criterion 2×10^7 times, with no errors.

6 The results

Theorem 6 (Block length). *If $\binom{n}{2}$ is a block of k primes and $n \notin \{4, 6, 15, 21, 715\}$, then $k \geq 1924$. This relies on Dusart’s theorem and the computation.*

Starts $\geq X_0$: §4. Starts $< X_0$: every block of ≤ 1923 primes was checked by computer (§5).

Theorem 7 (Bounded verification). *There is no solution other than 4, 6, 15, 21, 715 with $n \leq 10^{9614}$ (relying on Dusart’s theorem and the computation), and none with $n \leq 10^{12}$ using computation alone.*

Theorem 8 (Small starting primes). *No block starting at a prime below 100 and ending at a prime $\leq 4 \cdot 10^9$ gives a new solution. In particular, $n(n-1) = r\#$ (the product of all primes up to r , the “714 and 715” question of Nelson, Penney and Pomerance [3]) has no new solution with $r \leq 4 \cdot 10^9$.*

Context. On the forum [2], D. Weisenberg’s independent search reached $n > 10^{500}$. D. A. Corneth showed on OEIS A280992 [4] that any new solution has a prime factor larger than 17389. That covers every block of primes up to 17389, i.e. $n < 10^{3742}$. The bound 10^{9614} goes beyond both.

7 What Lean certifies

Lean is a proof assistant: a program that checks every logical step of a proof, down to the basic axioms. The balance inequality for consecutive-prime blocks is proved in Lean 4 with the Mathlib library, stated with the same definitions as the official formal statement of the problem:

`n.choose 2 =  $\prod$  i  $\in$  Finset.Ico a b, Nat.nth Nat.Prime i.`

“No sorry” means no step was left unproved. “No native_decide” means no step relied on trusting compiled code. Lean does *not* certify Dusart’s theorem or the computer searches.

8 Why the full problem looks hard

Random model. Each way of splitting the block’s primes into two teams determines one residue class for n modulo $2P$, which gives 2^k candidates in all. A solution needs the smallest candidate to be tiny, about $\sqrt{2P}$. If the candidates behaved like random numbers, the smallest would be about $2P/2^k$, which is vastly bigger than $\sqrt{2P}$ for long blocks. An exhaustive computation for blocks starting at 2 or 3 (up to $r = 97$) matches this random prediction closely. This is strong evidence that no further solutions exist, but it is not a proof.

Why standard tools fail.

- *abc conjecture.* It settles similar questions when numbers contain repeated prime factors. Here every prime appears once, so abc gives nothing.
- *Size arguments.* The equation $|2 \prod S - \prod T| = 1$ sits exactly at the smallest possible gap between distinct integers, so no size estimate can rule it out.
- *Fourier analysis.* The 2^k candidates are too few to be forced to “spread out” at the needed resolution.

9 Exercises

Exercise 1. Why can 7 never divide both n and $n - 1$?

Exercise 2. For $n = 21$, find E , O , A and the teams S and T .

Exercise 3. Use the test of §1.1 to decide whether $3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$ is of the form $\binom{n}{2}$.

Exercise 4. Why does a block starting near 10^{10} need so many primes?

Exercise 5. What is a “false positive” of the filter, and how is it caught?

Solutions

1. If $7 \mid n$ and $7 \mid n - 1$, then 7 divides their difference, which is 1. Impossible.
2. $E = 20$, $O = 21$, $A = 10$. $S = \{2, 5\}$ (since $A = 2 \cdot 5$) and $T = \{3, 7\}$ (since $O = 3 \cdot 7$). Check: $2A = 20 = O - 1$.
3. $P = 15015$, $8P + 1 = 120121$, and $346^2 = 119716 < 120121 < 120409 = 347^2$. Not a square, so no.
4. Consecutive primes there are about 23 apart, so each step raises r/p by only a factor of about $1 + 2.3 \times 10^{-9}$. Reaching $(r/p)^{k/2} \approx 2$ needs an enormous k .
5. A block whose $8P + 1$ is not a square but happens to be a square modulo all 48 filter primes. It is caught by the exact integer square-root check. None occurred.

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References

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- [3] C. Nelson, D. E. Penney and C. Pomerance, 714 and 715, *J. Recreational Math.* **7**(2) (1974), 87–89.
- [4] OEIS Foundation, Sequence A280992, <https://oeis.org/A280992>.